

Analytical

22-07-2026

1) Regular Expression for following binary string languages

a) Only 0's

RE: 0^*

b) All binary strings

RE: $(0+1)^*$

c) All binary strings except empty string

RE: $(0+1)(0+1)^*$

d) Begins with 1 and ends with 1

RE: $1(0+1)^*1$

e) Ends with 00

RE: $(0+1)^*00$

f) Contains at least three 1's

RE: $0^*10^*10^*10^*(0+1)^*$

g) Contains at least three consecutive 1's

RE: $(0+1)^*111(0+1)^*$

h) RE: $(0+1)^*110(0+1)^*$ contains substring 110

i) This cannot be expressed

j) $1^*01+01^*(1+01^*)^*$

k) $(0+1)(0+1)0(0+1)^*$

Number of 0's is a multiple of 3.

$$RE: (0+1)(0+1)^2(0+1)^*$$

2) No. of 0's is start and ends with same character

$$RE: (0+1)(0+1)^*(0+1)$$

3) odd length

$$RE: ((0+1)(0+1))^*(0+1)$$

4) Starts with 0 and has odd length

$$RE: 0((0+1)(0+1))^*(0+1) + 1((0+1)(0+1))^*$$

5) Length is at least 1 and at most 3

$$RE: (0+1) + (0+1)(0+1) + (0+1)(0+1)(0+1)$$

2) a) Language starting and ending with a & having any combination of b's in b/w

Regular expression:

$$ab^*a$$

Examples:

• aa • abba
• aba • abbaa

b) Language starting with a but not having consecutive b's

Regular expression:

$$a(a+ba)^*$$

Examples:

• a • ababa
• aa
• aba
• abaa

c) language accepting all strings in which any no. of a's & b's & c's

regular expression:
 $a^*b^*c^*$

examples:

• ϵ • aaabbbccc
• aaa • bbbccc
• aaabbb

d) language over $\Sigma = \{0\}$ having even length of string

regular expression:
 $(00)^*$

examples:

• ϵ • 0000
• 00 • 000000

e) language having at least one 0 and at least one 1

regular expression:

$(0+1)^*0(0+1)^*1(0+1)^* + (0+1)^*1(0+1)^*0(0+1)^*$

examples:

• 01
• 10
• 101
• 0011
• 1100

3) RE to FA

a) a

$\rightarrow q_0 \xrightarrow{a} (q_1)$

Accept state: q_1

b) ab

$\rightarrow q_0 \xrightarrow{a} q_1 \xrightarrow{b} (q_2)$

Accept state: q_2

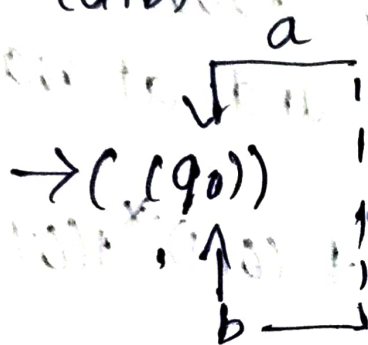
c) $RE = (a+b)$

$\rightarrow q_0 \xrightarrow{a} (q_1)$

$\rightarrow q_0 \xrightarrow{b} (q_1)$

Accept state: q_1

d) $RE = (a+b)^*$



Accept state q_0

e) $RE = 1(0+1)^*0$

$\rightarrow q_0 \xrightarrow{1} q_1$

$\rightarrow q_1 \xrightarrow{0} q_2$

$\rightarrow q_1 \xrightarrow{1} q_1$

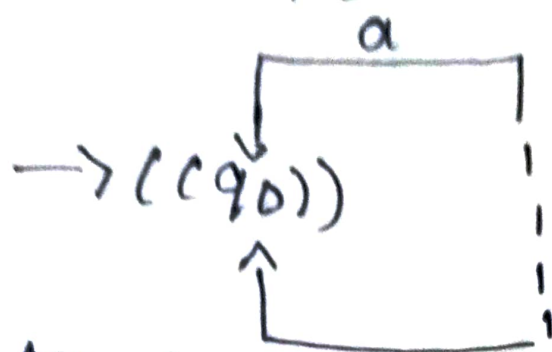
$\rightarrow q_2 \xrightarrow{0} q_2$

$\rightarrow q_2 \xrightarrow{1} q_1$

Accept state q_2

4) convert RE to FA

a) $a^* + b^* + c^*$



$q_0 \xrightarrow{b} q_1$

$q_1 \xrightarrow{b} q_1$

$q_0 \xrightarrow{c} q_2$

$q_2 \xrightarrow{c} q_2$

Accept states: q_0, q_1, q_2

b) $(aa)^*(bb)^* + a(aa)^*b(bb)^*$

$\rightarrow q_0 \xrightarrow{a} q_1$

$\rightarrow q_1 \xrightarrow{a} q_0$

$\rightarrow q_0 \xrightarrow{b} q_2$

$\rightarrow q_2 \xrightarrow{b} q_0$

Accept state: q_0

c) $RE = (1+01)^*00(1+10)^*$

$\rightarrow q_0 \xrightarrow{1} q_0$

$\rightarrow q_0 \xrightarrow{0} q_1$

$\rightarrow q_1 \xrightarrow{1} q_0$

$\rightarrow q_1 \xrightarrow{0} q_2$

$\rightarrow q_2 \xrightarrow{1} q_2$

$\rightarrow q_2 \xrightarrow{0} q_2$

$\rightarrow q_3 \xrightarrow{1} q_2$

Accept state: q_2

$$d) (0+1)(1+1)(1+2)^*0(2+1)^*$$

FA :

$$\rightarrow q_0 \xrightarrow{0|1} q_1$$

$$\rightarrow q_1 \xrightarrow{1|1} q_2$$

$$\rightarrow q_2 \xrightarrow{1|2} q_2$$

$$\rightarrow q_2 \xrightarrow{0} q_3$$

$$\rightarrow q_3 \xrightarrow{1|2} q_3$$

Accept state: q_3

$$e) (cat+bt+c)^*(a+bt+c+d)$$

$$\rightarrow (q_0) \xrightarrow{a|b|c} q_1$$

$$q_1 \xrightarrow{c} q_0$$

$$q_0 \xrightarrow{a|b|c} q_2$$

Accept states: q_0, q_2

5) construct table

(a)

state	a	b
$\rightarrow q_1$	q_2	q_1
q_2	q_2	q_3
q_3	q_4	q_1
$*q_4$	q_2	q_3

start state: q_1

final state: q_4

b)

state	a	b
→ q ₁	q ₄	q ₃
q ₂	q ₃	q ₃
*q ₃	q ₃	q ₃
q ₄	q ₄	q ₄

Start state: q₁
Final state: q₃

c)

state	a	b
→ q ₁	q ₂	q ₄
q ₂	q ₃	q ₅
*q ₃	q ₃	q ₆
q ₄	q ₅	q ₇
q ₅	q ₆	q ₇
*q ₆	q ₆	q ₇
q ₇	q ₇	q ₇

Start state: q₁
Final state: q₃, q₆

d)

state	a	b
→ *q ₁	q ₁	q ₂
q ₂	q ₁	q ₃
*q ₃	q ₁	q ₃

e)

state	a	b
→ *q ₁	q ₂	q ₃
q ₂	q ₄	q ₅
*q ₃	q ₆	q ₇
q ₄	q ₈	q ₈
*q ₅	q ₈	q ₈
q ₆	q ₈	q ₈
q ₇	q ₈	q ₈
q ₈	q ₈	q ₈

start state: q₁
final state: q₁, q₃, q₅

f) -

state	a	b
→ q ₁	q ₂	q ₁
*q ₂	q ₃	q ₂
q ₃	q ₁	q ₃

start: q₁
final: q₂

6) Inflectional & derivational morphology

Word	Inflectional	Derivational
a) work	work → works, worked	work → worker, workplace
b) Red	Red → redder, reddest	red → reddish
c) Tree	tree → trees	tree → treeless
d) soft	soft → softer, softest	soft → sotten
e) Happy	happy → happier, happiest	happy → happiness

1) ^{a)} New York is the most densely populated city in the United States

stemmed words:

• New → New

• York → York

• is → is

• the → the

• most → most

• densely → dens

• populated → popul

• city → citi

• in → in

• the → the

• United → unit

• States → state

lemmatized words:

- New → new
- York → york
- is → be
- the → the
- most → most
- densely → densely
- populated → populate
- city → city
- in → in
- the → the

- united → united
- states → state

b) My mother had a beautiful singing voice

stemmed words

- My → my
- mother → mother
- had → had
- a → a
- beautiful → beautiful
- sing → sing
- voice → voice

lemmatized words

- My → my
- mother → mother
- had → have
- a → a
- beautiful → beautiful
- singing → sing
- voice → voice

8) (a) Email Address

$^1[A-Za-z0-9._]+\@[A-Za-z0-9]+\.[A-Za-z]{2,4}$

(b) phone number (10 digits)

$^1[0-9]{10}$

(c) Date (DD/MM/YYYY)

$^1\d{2}/\d{2}/\d{4}$

(d) Time (HH:MM)

$^1[0-1]?[0-9]{1,2}:[0-5][0-9]$

(e) IP Address

$^1(\d{1,3}\.){3}\d{1,3}$

(f) URL

$^1https?://[A-Za-z0-9._]+\.[A-Za-z]{2,4}^*$

(g) Credit Card number (16 digits)

$^1[0-9]{16}$

(h) pincode (India)

$^1[0-9]{6}$

(i) password

$^1(?=.*[A-Za-z])(?=.*\d)\S{4,}$

9) Sol:

Using Zipf's law:

$\therefore [\text{Rank} \times \text{Frequency} = \text{constant}]$

Given:

Rank, = 4

frequency, = 600

frequency, = 300

$$4 \times 600 = \text{Rank,} \times 300$$

$$2400 = \text{Rank,} \times 300$$

$$\boxed{\text{Rank,} = 8}$$

10) Sentence :
The only thing we have to tear is tear

Count :

• Word Tokens (Total words) = 10

• the, only, thing, we, have, to, tear, is, it, itself

• Word Type (Unique words) = 9

• the, only, thing, we, have, to, tear, is, it

$$\text{Tokens} : \text{Types} = 10 : 9$$

11) using the meaning-Rank relationship:

$$r_1 \propto \frac{1}{\sqrt{r_2}}$$

Given:

• $r_1 = 1600$

• $r_2 = 400$

$$m_1 : m_2 = \frac{1}{\sqrt{1600}} : \frac{1}{\sqrt{400}} = \frac{1}{40} : \frac{1}{20} = 1 : 2$$

$$\therefore m_1 : m_2 = 1 : 2$$

12) Sol:

$$V = K N^\beta$$

By using Heap's Law

$$\bullet N = 12000$$

$$\bullet K = 3.71$$

$$\bullet \beta = 0.69$$

$$V = 3.71 \times (12000)^{0.69}$$

$$V = 3.71 \times 651.6$$

$$V \approx 2418$$

Ans

$\therefore$ unique words ≈ 2418